

Curried Functional Spaces

Roman Maksimovich

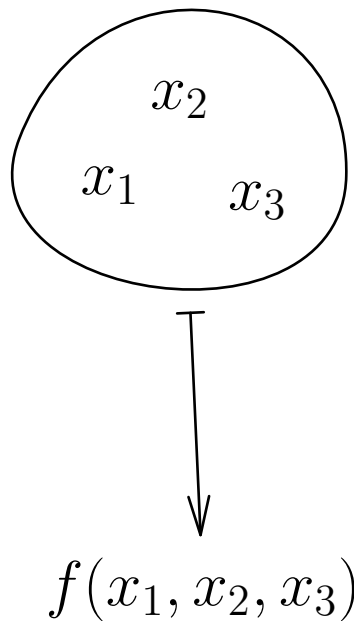
Hong Kong University of Science and Technology

May 10, 2026

Premise

Multivariable functions

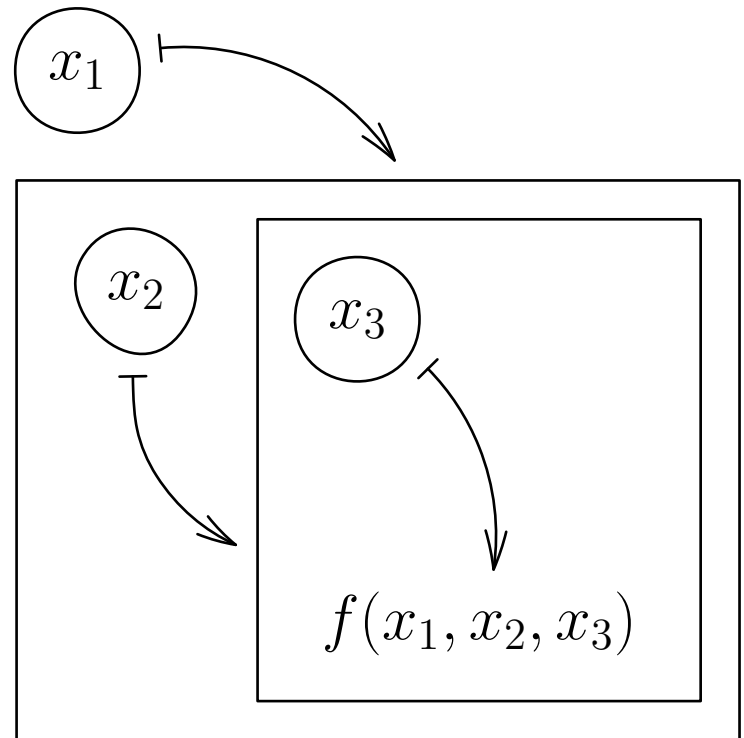
$$\mathbb{R}^n \rightarrow \mathbb{R}$$



Multivariate calculus

Curried functions

$$\mathbb{R} \rightarrow (\mathbb{R} \rightarrow (\dots (\mathbb{R} \rightarrow \mathbb{R})))$$

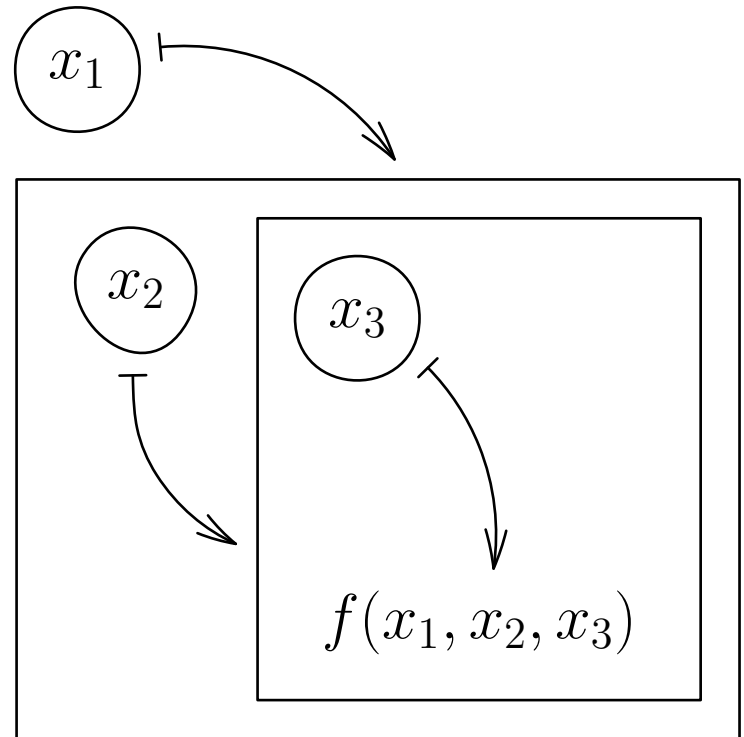


λ -calculus, functional programming

Premise

Curried functions

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λ -calculus,
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Multivariate calculus

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n -ary curried functions:

1. $C_0(\mathbb{R}) := \mathbb{R}$

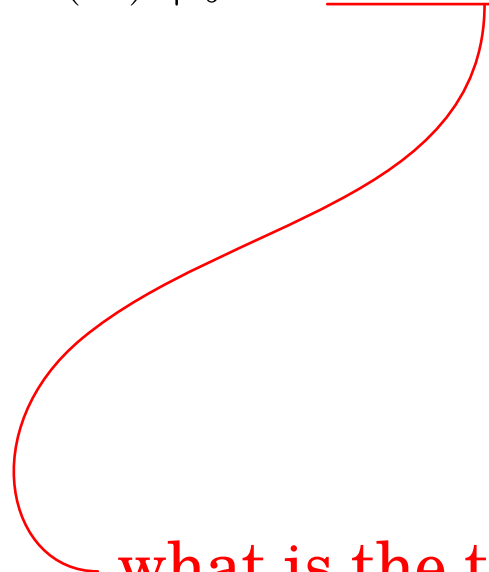
2. $C_{n+1}(\mathbb{R}) := \{f: \mathbb{R} \rightarrow C_n(\mathbb{R}) \mid f \text{ is continuous}\}$

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what is the topology on $C_n(\mathbb{R})$?

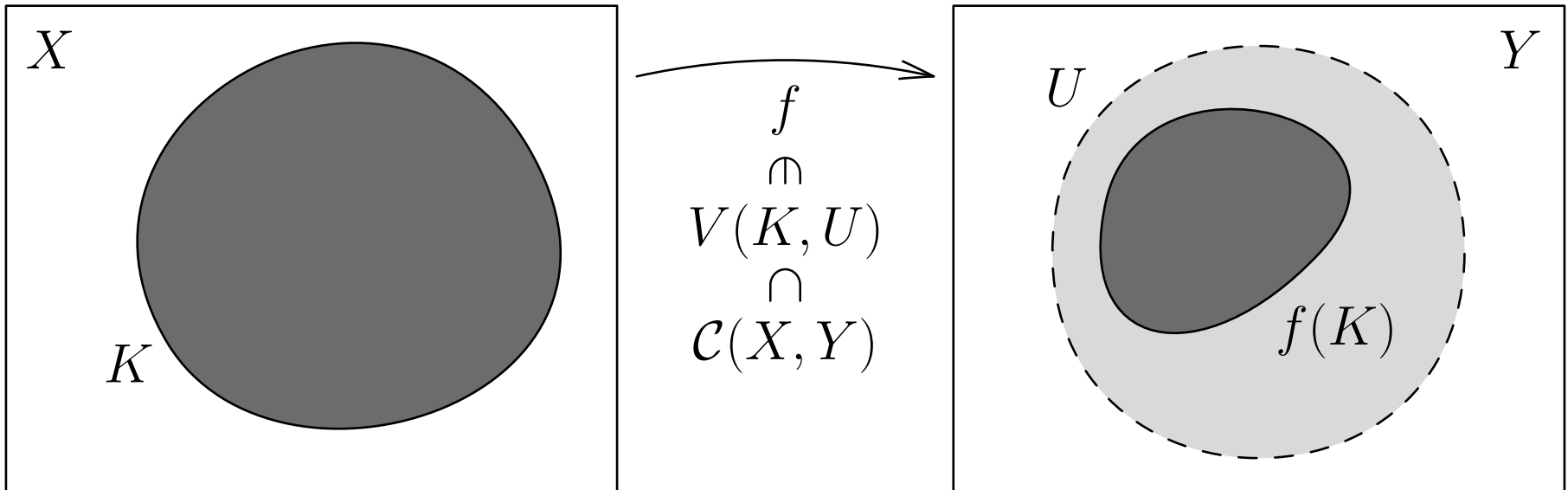
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compact-open topology: $C_n(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R}))$



Topological Vector Spaces

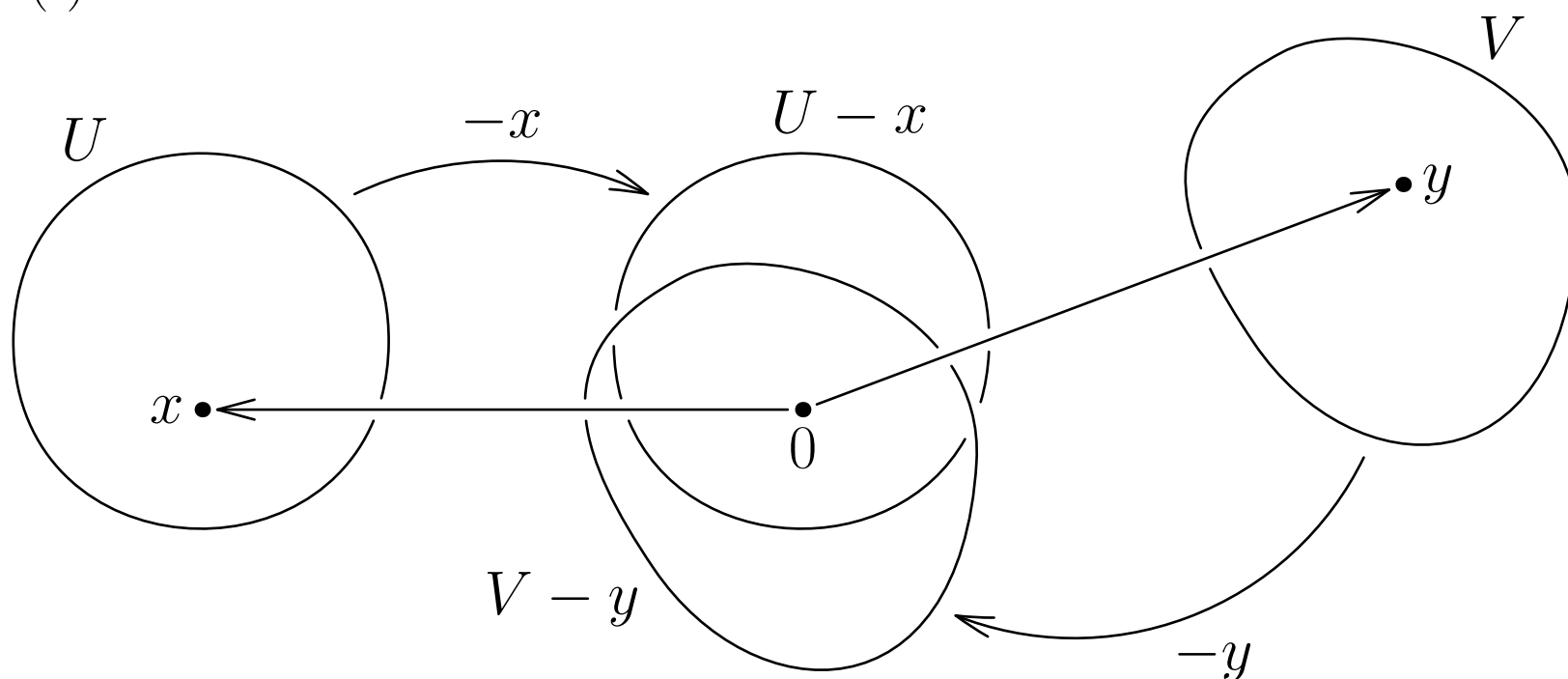
Remark: If $f_1, f_2 \in C_n(\mathbb{R})$, then $\alpha f_1 + \beta f_2 \in C_n(\mathbb{R})$.

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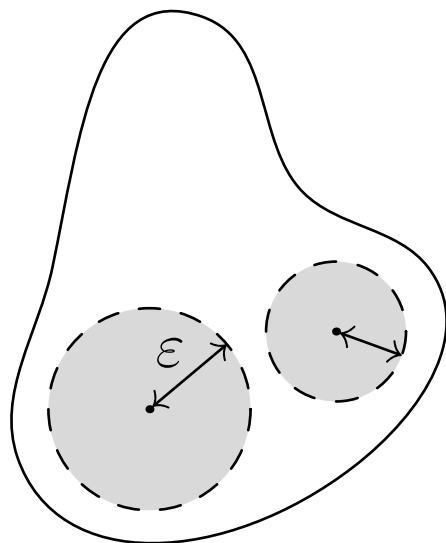
Definition: X is a **TVS** $\Leftrightarrow X$ has **topological** and **vector** structure, and they are *compatible*:

1. $(+): X \times X \rightarrow X$ is continuous;
2. $(\cdot): \mathbb{R} \times X \rightarrow X$ is continuous.

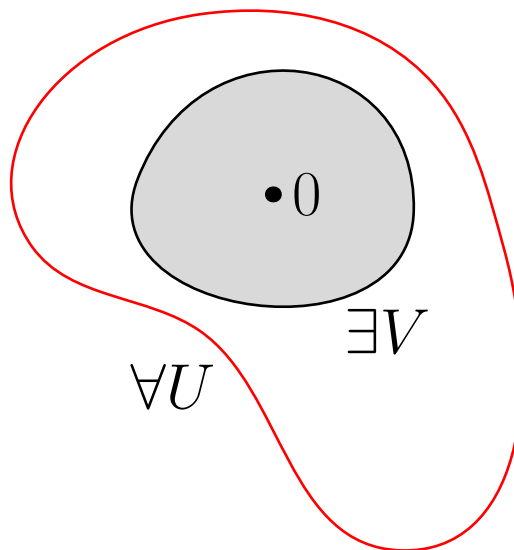


Properties of Topological Vector Spaces

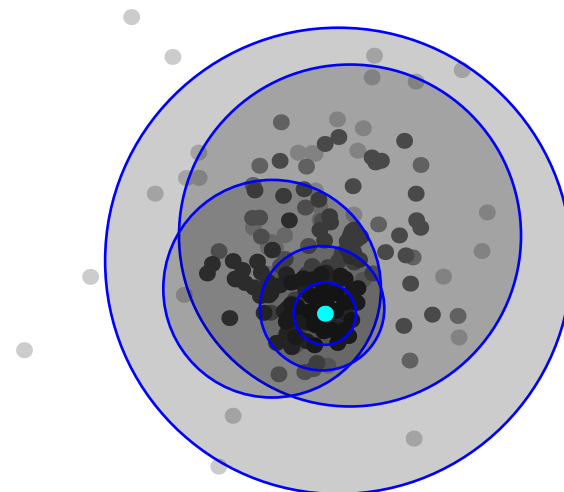
Metrizability



Local convexity



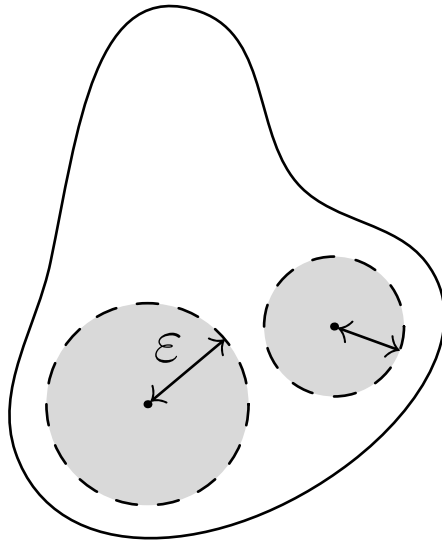
Completeness



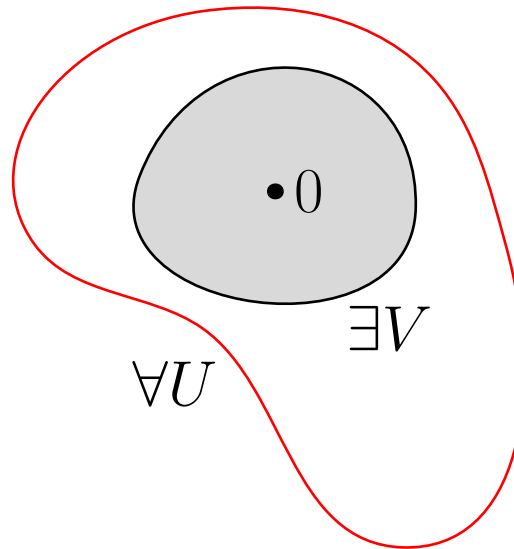
Fréchet space

Properties of Topological Vector Spaces

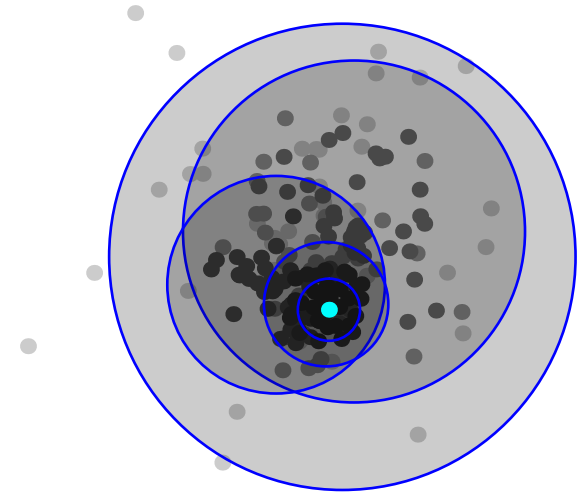
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Curried function spaces $C_n(\mathbb{R})$

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n -ary curried functions:

1. $C_0(\mathbb{R}) := \mathbb{R}$;
 2. $C_{n+1}(\mathbb{R}) := \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, with the compact-open topology.
-

Results: for all $n \in \mathbb{N}$,

1. $C_n(\mathbb{R})$ is a Hausdorff topological vector space;
 2. There is an isomorphism $\gamma: \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \leftrightarrow C_n(\mathbb{R})$;
 3. $C_n(\mathbb{R})$ is a Fréchet space.
-

$\Rightarrow$ Theorems from functional analysis apply:

1. Uniform boundedness principle;
2. Open mapping theorem;
3. ...

Differentiable functions

$D_n^m(\mathbb{R})$ — the space of n -ary m times differentiable functions:

1. $D_0^m(\mathbb{R}) := \mathbb{R};$

2. $D_n^0(\mathbb{R}) := C_n(\mathbb{R});$

3. If $n, m > 0$, then $D_n^m(\mathbb{R})$ consists of all such $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$, that there is $f' \in D_n^{m-1}(\mathbb{R})$ with

$$\frac{f(x+h) - f(x)}{h} \xrightarrow{h \rightarrow 0} f'(x).$$

Differentiable functions

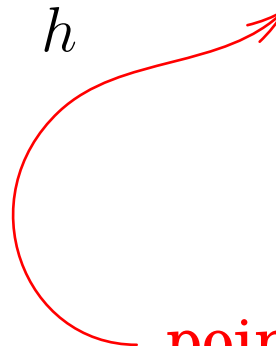
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pointwise convergence!

$$f_k \xrightarrow{k \rightarrow \infty} f \Leftrightarrow f_k(x_1) \dots (x_n) \xrightarrow{k \rightarrow \infty} f(x_1) \dots (x_n)$$

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The topology on D_n^m is *coarsest* such that the maps

$$\begin{array}{ccc} i: D_n^m(\mathbb{R}) \rightarrow \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), & d: D_n^m(\mathbb{R}) \rightarrow D_n^{m-1}(\mathbb{R}) \\ f \mapsto f & f \mapsto f' \end{array}$$

are continuous. In particular, $f_k \rightarrow f \Rightarrow f'_k \rightarrow f'$.

Differentiable functions

$D_n^m(\mathbb{R})$ — the space of n -ary m times differentiable functions:

Results: for all $m, n \in \mathbb{N}$,

1. $D_n^m(\mathbb{R})$ is a Hausdorff topological vector space;
2. The inclusion map $j: D_n^{m+1}(\mathbb{R}) \rightarrow D_n^m(\mathbb{R})$ is continuous;
3. There is an isomorphism $\gamma: C^m(\mathbb{R}^n) \leftrightarrow D_n^m(\mathbb{R})$, given by

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)\dots(x_n) = \tilde{f}(x_1, x_2, \dots, x_n).$$

4. $D_n^m(\mathbb{R})$ is a Fréchet space.

References

- [1] N. Bourbaki, *General Topology, Part I*. 1966
- [2] N. Bourbaki, *General Topology, Part II*. 1966
- [3] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002
- [4] W. Rudin, *Principles of Mathematical Analysis*. 1976
- [5] W. Rudin, *Functional Analysis*. 1976